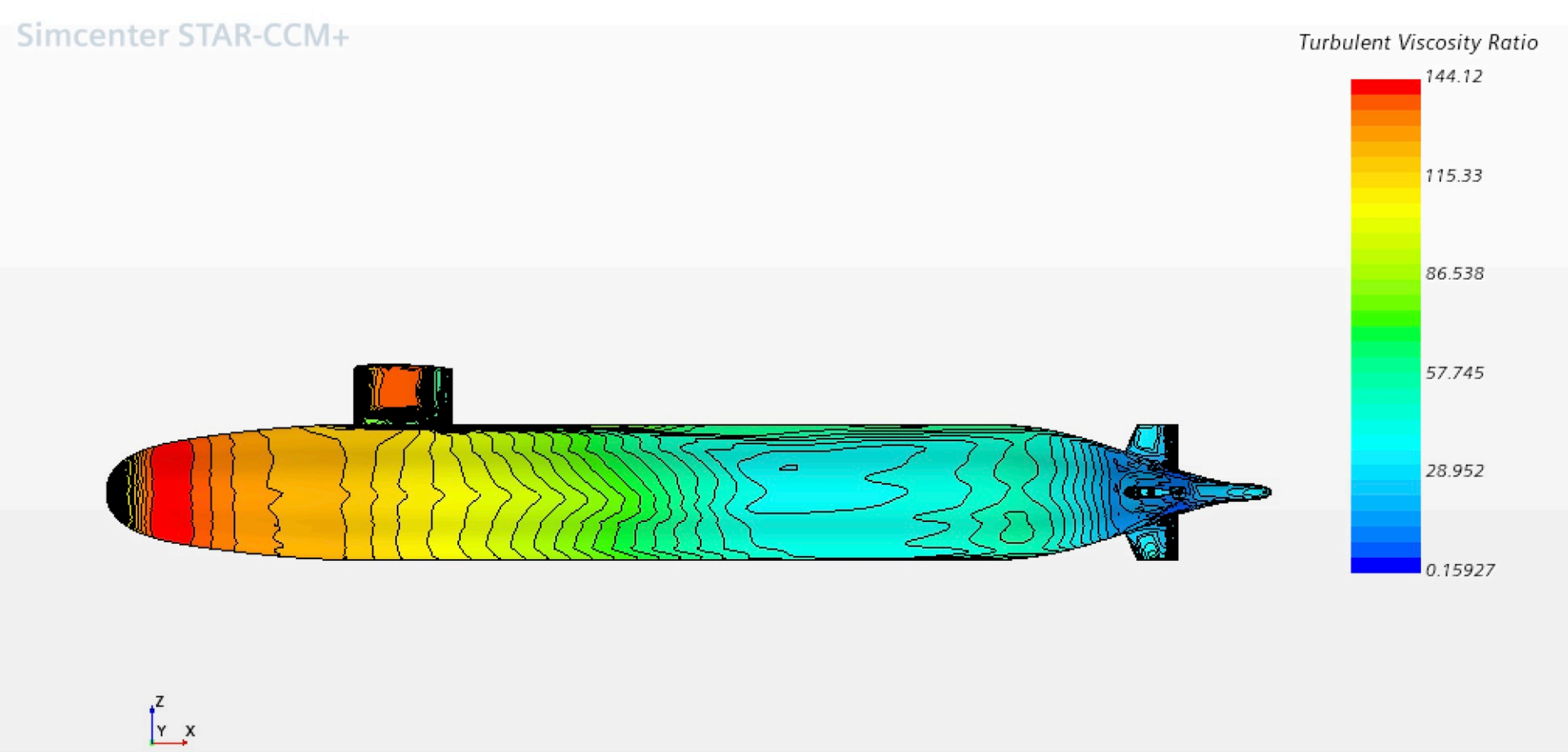
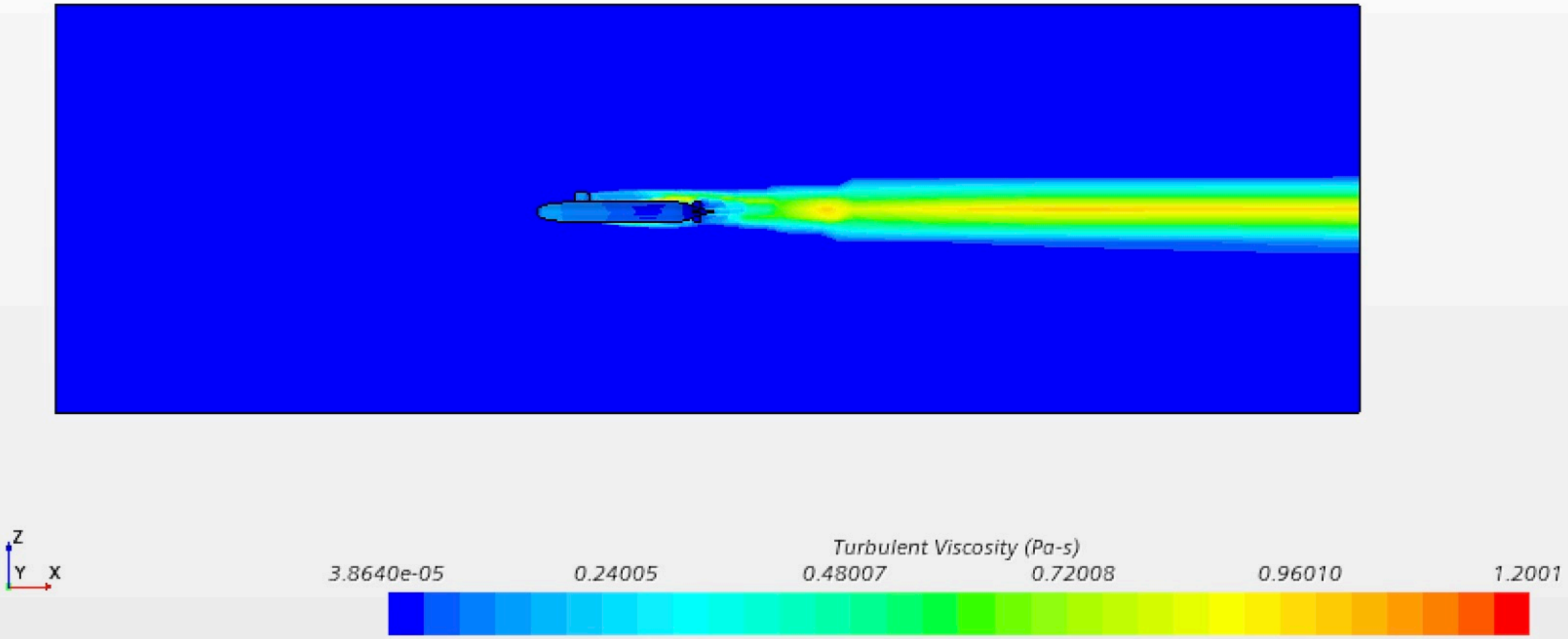
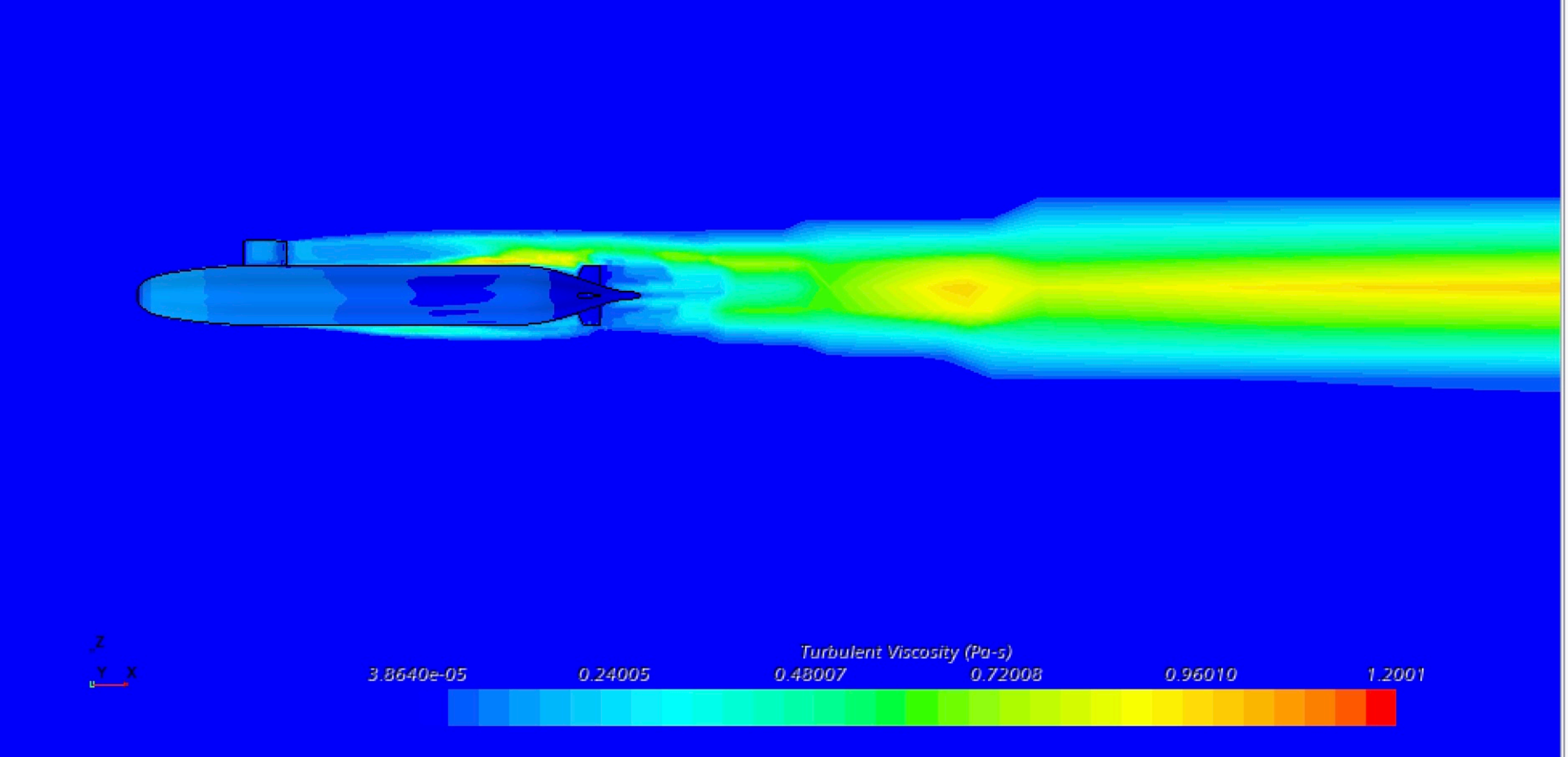
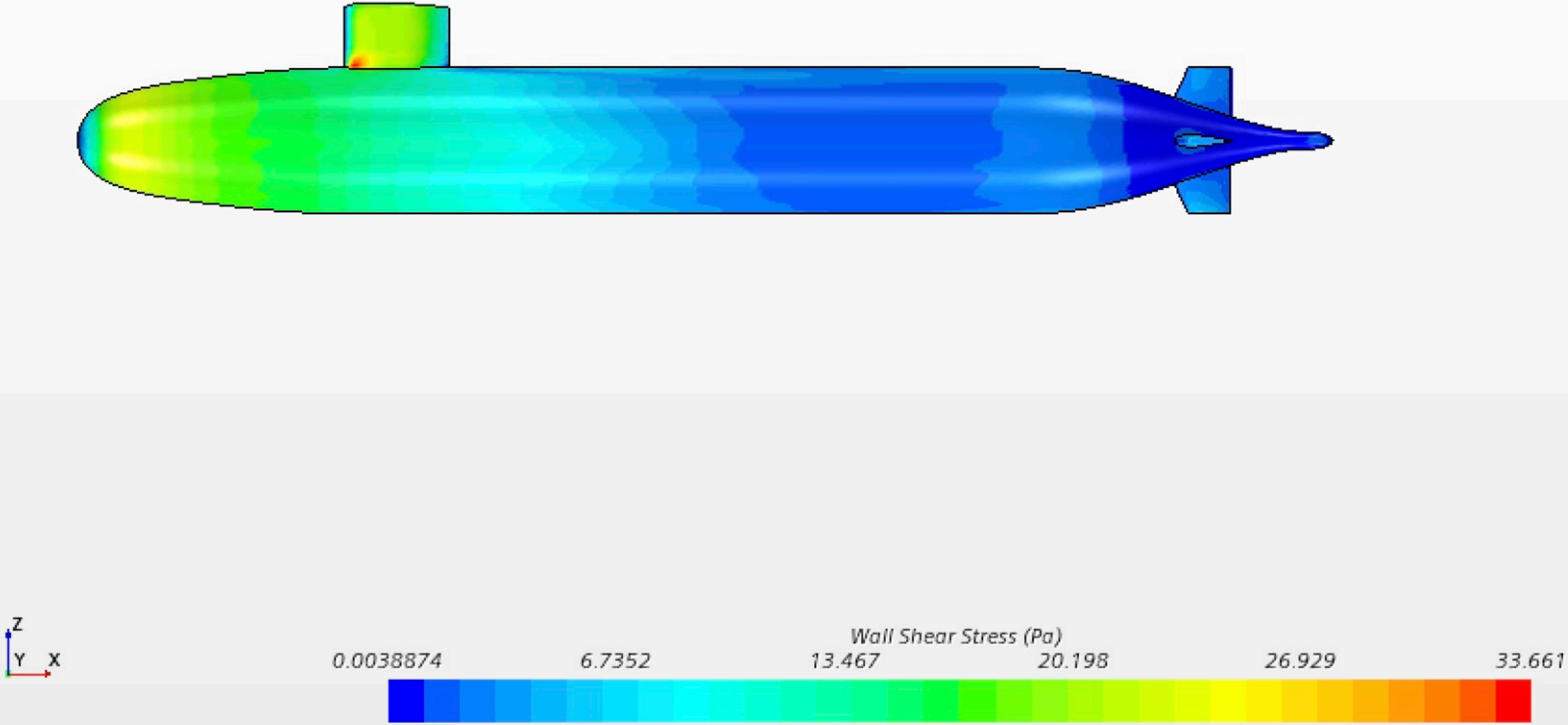
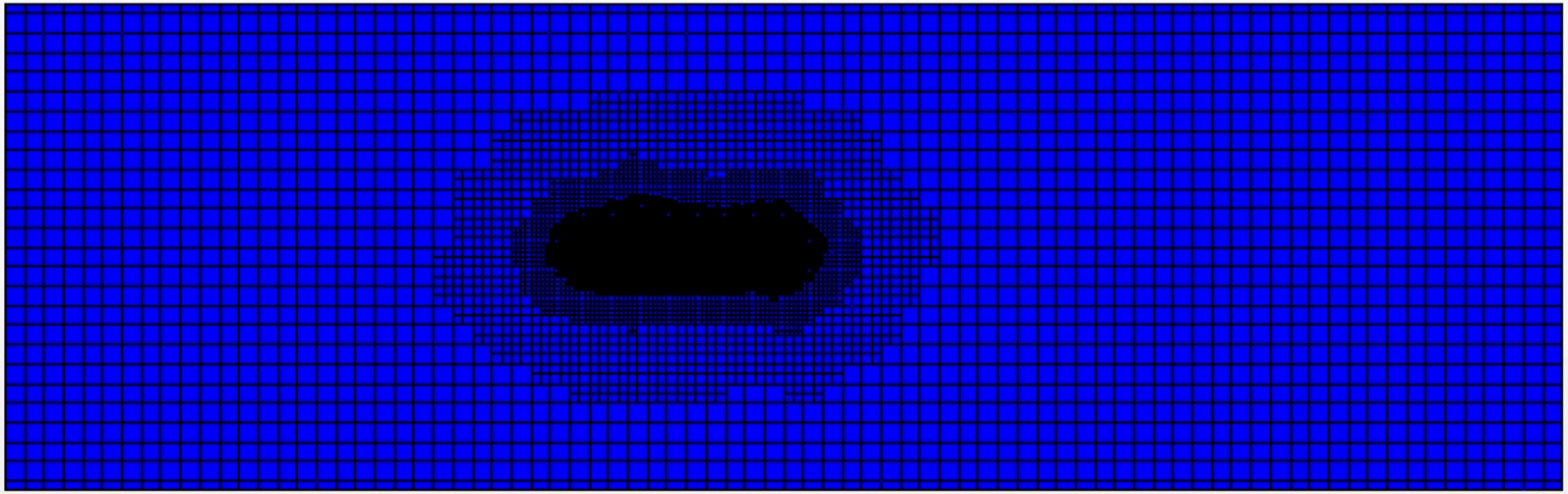

Simcenter STAR-CCM+
Turbulent Viscosity Ratio
144.12
115.33
86.538
57.745
28.952
0.15927
Z
Y
X

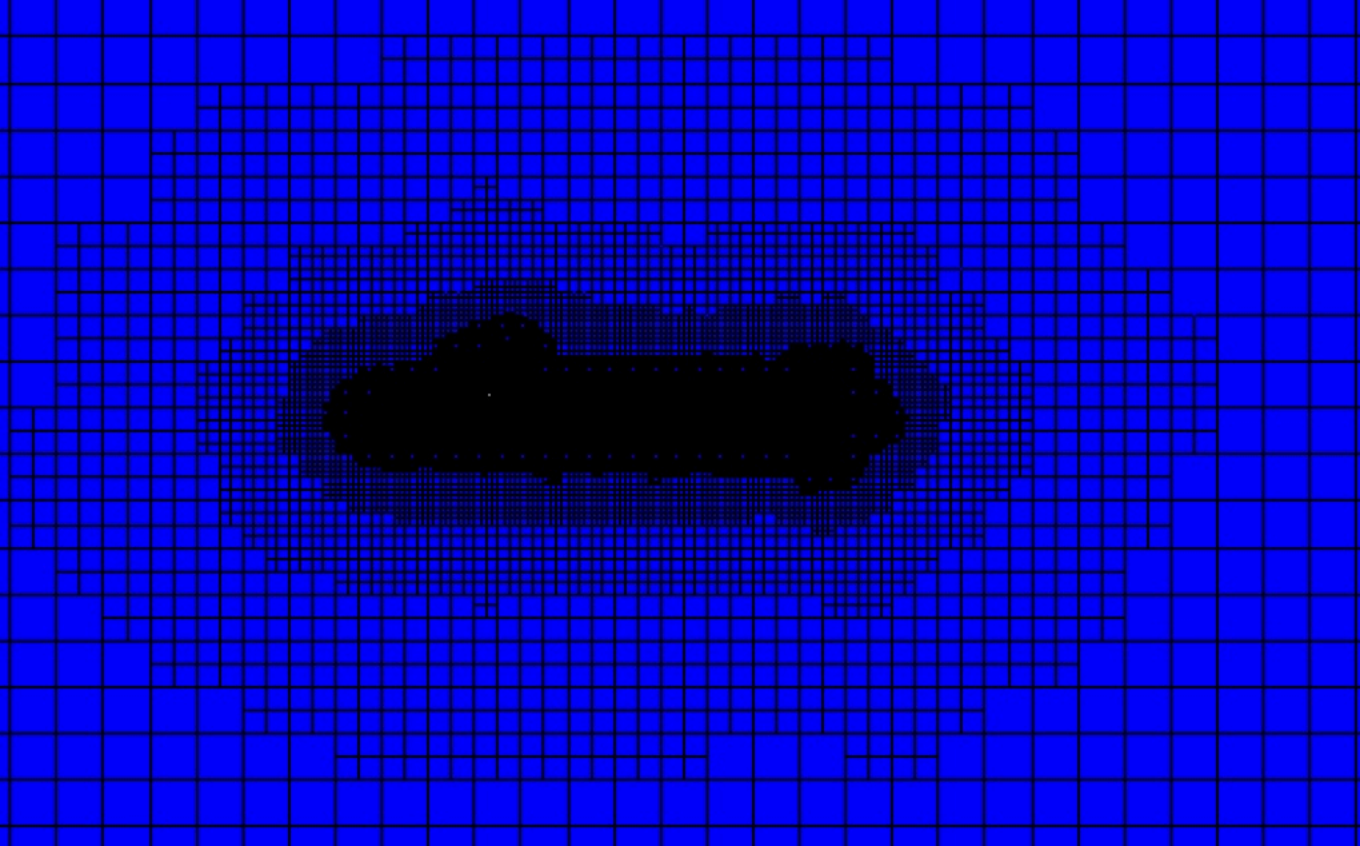

Z
Y
X
Turbulent Viscosity (Pa-s)
3.8640e-05
0.24005
0.48007
0.72008
0.96010
1.2001

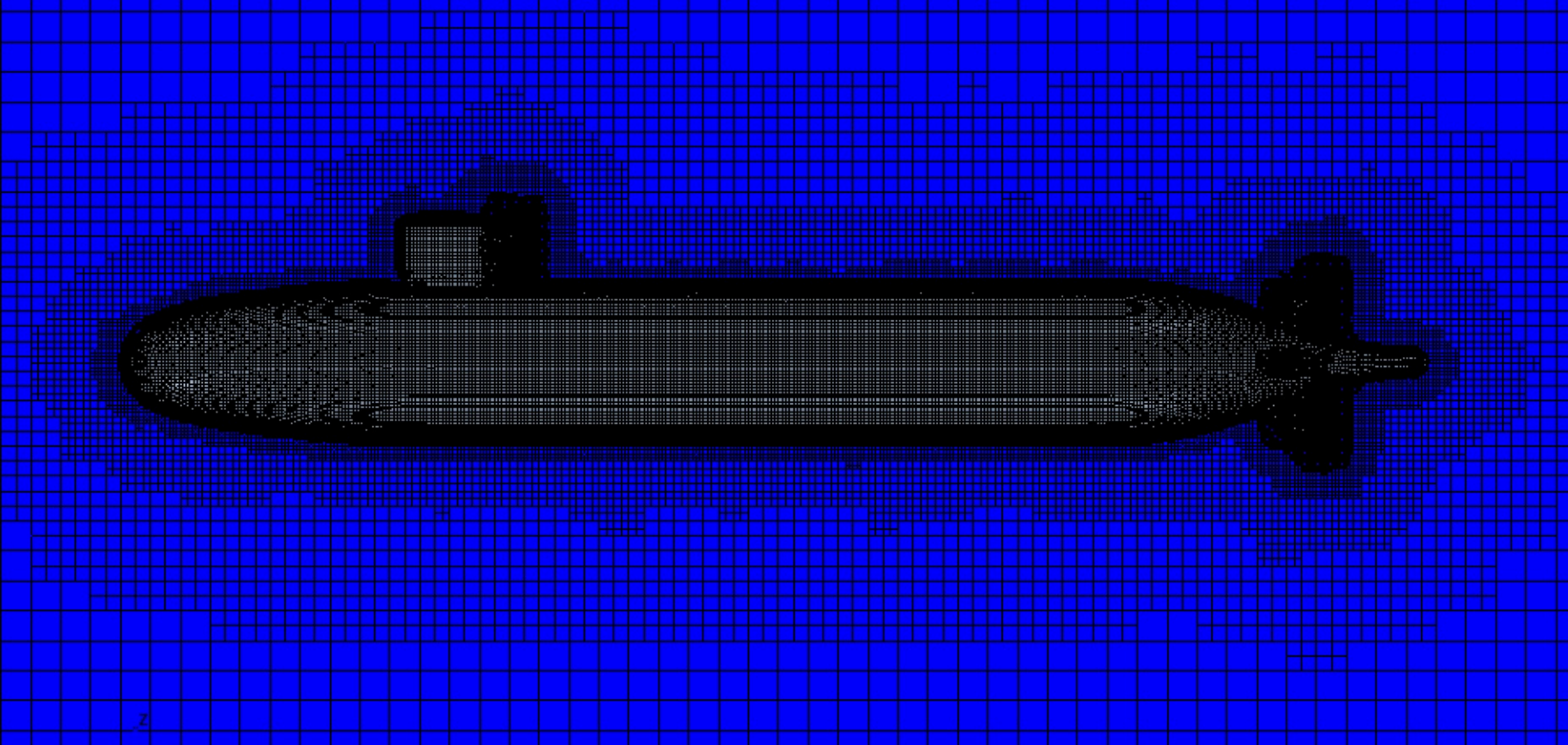

Z
Y X
Turbulent Viscosity (Pa-s)
3.8640e-05
0.24005
0.48007
0.72008
0.96010
1.2001

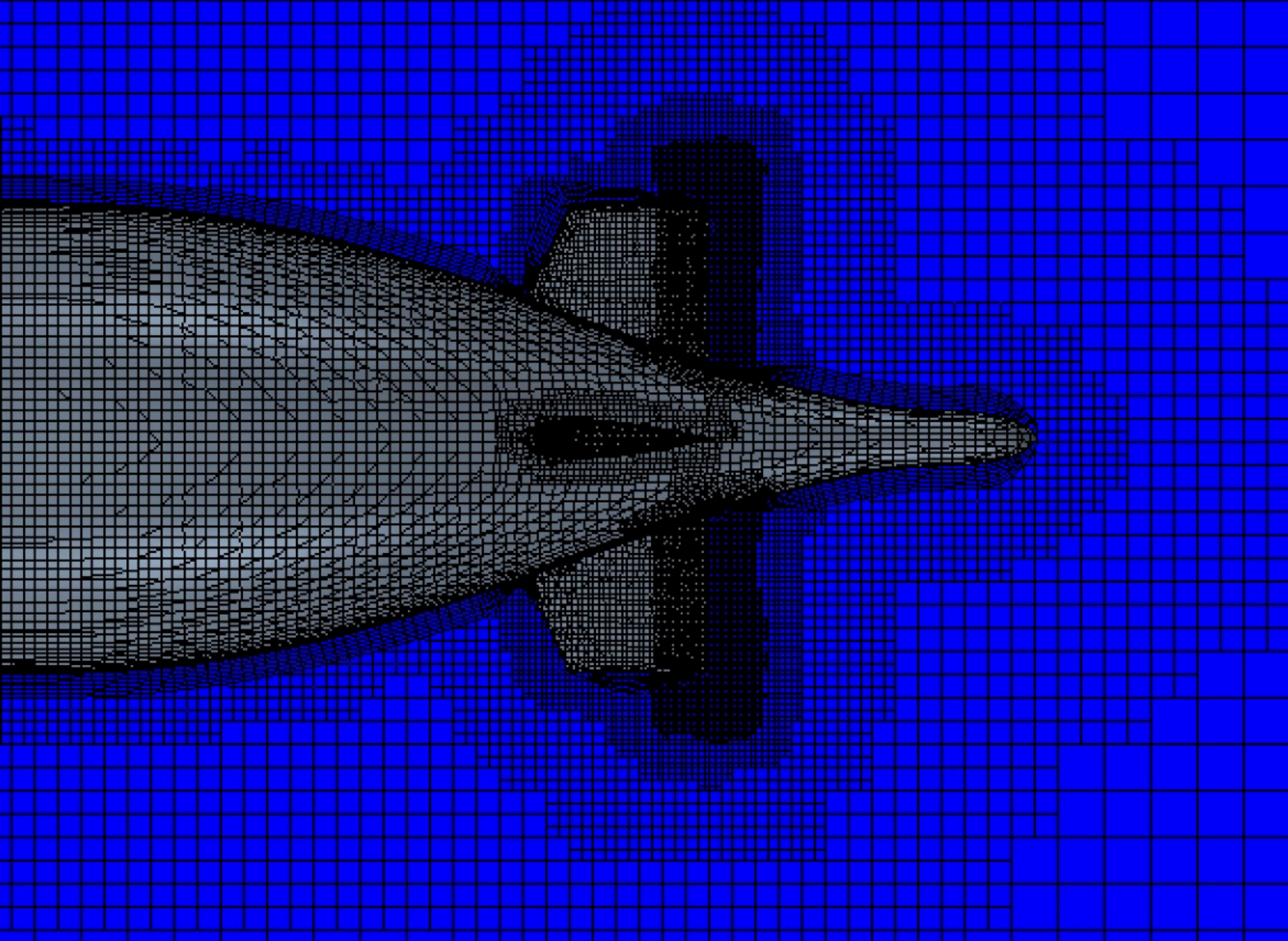

Z
Y
X
Wall Shear Stress (Pa)
0.0038874
6.7352
13.467
20.198
26.929
33.661

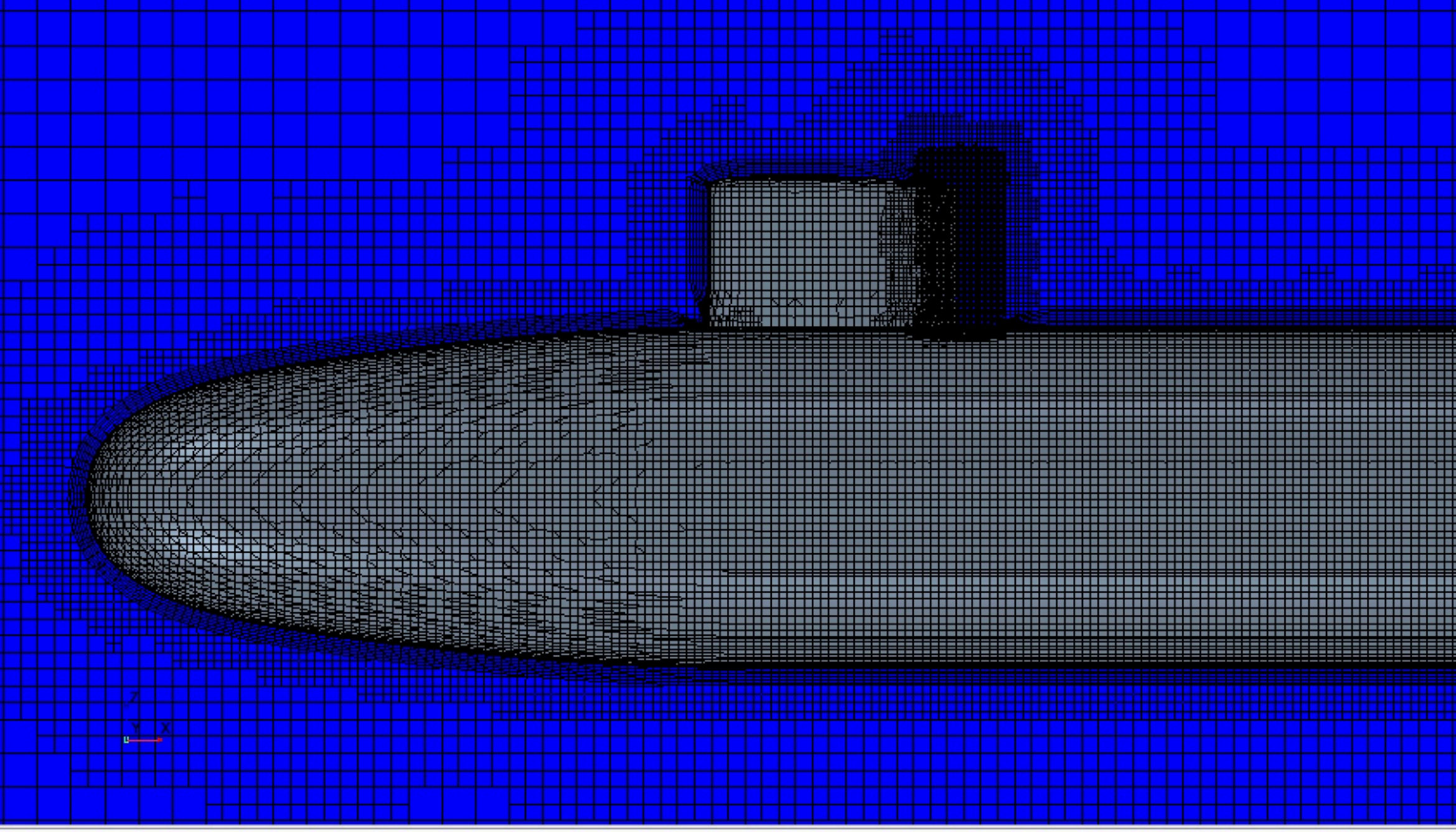


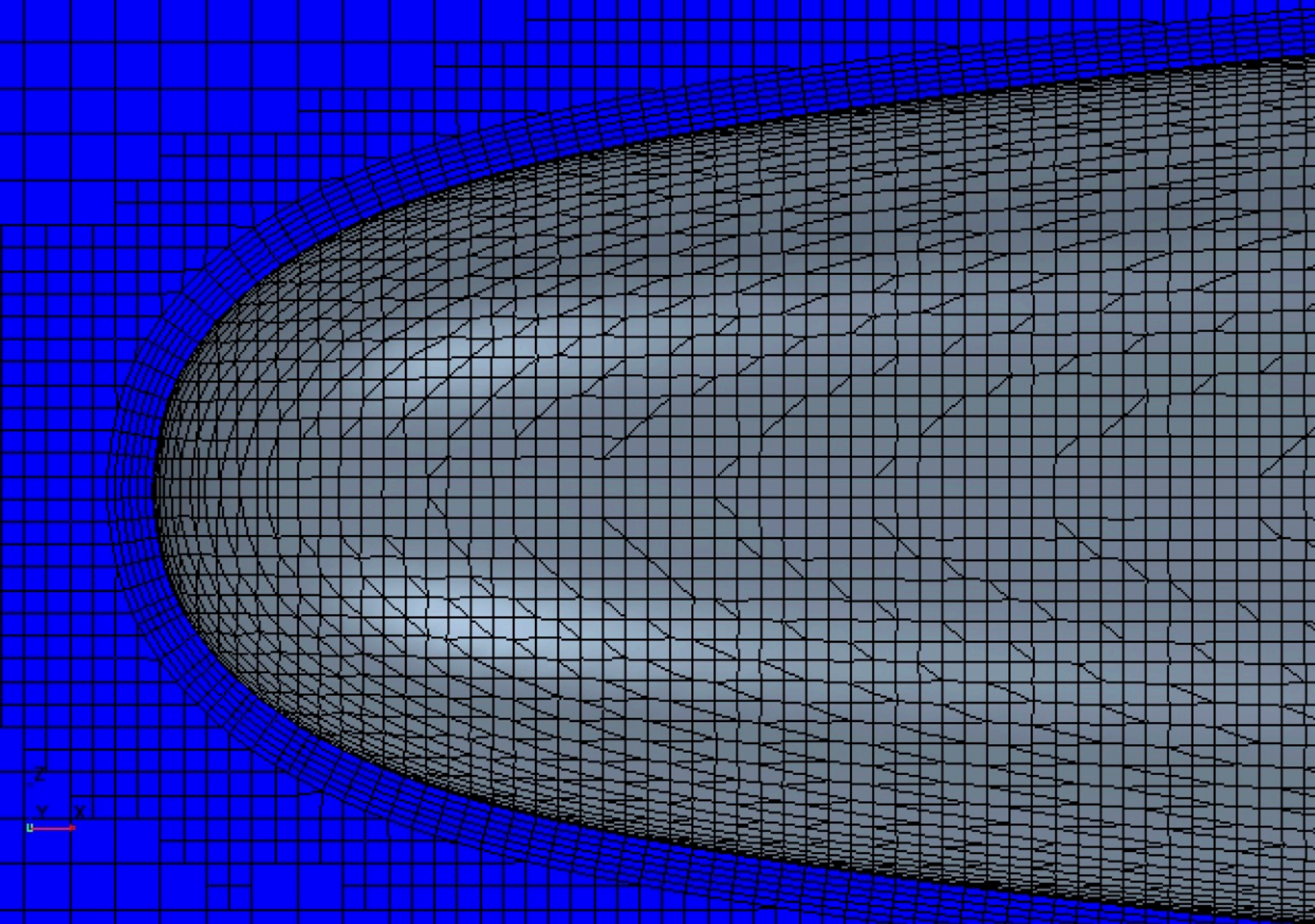
Z




Z




Z
Y
X


Z
Y
X

Summary Report: submarine

Session Summary

Date	Feb 22, 2026, 11:35:36 AM
Simulation	Z:\Star CCM +\submarine.sim
File size	1.4e+02 MB
Number of Partitions	2
Number of Restored Partitions	2

Software Summary

Version	BuildArch: win64 BuildEnv: intel18.3vc14-r8 PresentationVersion: 2020.2.1 ReleaseDate: Thu Jul 16 15:33:18 UTC 2020 ReleaseNumber: 15.04.010
MPI Version	IBM Platform MPI-9.1.4.4

Hardware Summary

Hosts	Number Processes: 2 Rank[0]: VAMSHIKRISH8130 Rank[1]: VAMSHIKRISH8130

Simulation Properties

submarine

Continua

Continua	2

Parts Meshes

Regions	[Region]
Interfaces	[]
Tags	[]

Physics 1

Regions	[Region]
Interfaces	[]
Point Sets	[]
Active	true
Tags	[]

Models

All y+ Wall Treatment

Iterative Ustar	false

Gradients

Limiter Method	Venkatakrishnan
Custom Accuracy Level Selector	2.0
Verbose	false
Least-Squares Quality Criterion	true
Flat Cells Curvature Criterion	true
Cell Skewness Criterion	true
Chevron-Cell Criterion	true
Least-Squares Tensor Minimum Eigenvalues Ratio	0.1
Normalized Flat Cells Curvature Factor	1.0
Maximum Safe (Positive) Skewness Angle (deg)	75.0
Minimum Unsafe (Positive) Skewness Angle (deg)	88.0
Use TVB Gradient Limiting	false
Acceptable Field Variation (Factor)	0.05

IAPWS-IF97 (Water)

K-Omega Turbulence

Liquid		
H2O	Database Material	H2O (Water) [Standard/Liquids]
	Tags	[]
Material Properties		
Critical Pressure	Method	IAPWS-IF97
IAPWS-IF97	Value	2.2064E7 Pa
Critical Temperature	Method	IAPWS-IF97
IAPWS-IF97	Value	647.096 K
Dynamic Viscosity	Method	Constant
Constant	Value	8.8871E-4 Pa-s
Molecular Weight	Method	IAPWS-IF97
IAPWS-IF97	Value	18.015154292499233 kg/kmol
Saturation Pressure	Method	IAPWS-IF97
IAPWS-IF97		
Saturation Temperature	Method	IAPWS-IF97
IAPWS-IF97		
Specific Heat	Method	IAPWS-IF97
IAPWS-IF97		
Speed of Sound	Method	IAPWS-IF97
IAPWS-IF97		
Thermal Conductivity	Method	Constant
Constant	Value	0.620271 W/m-K
Turbulent Prandtl Number	Method	Constant
Constant	Value	0.9
Reynolds-Averaged Navier-Stokes		
Segregated Flow	Positivity Rate Limit	0.2
	Flow Boundary Diffusion	true
	Unsteady Flux Dissipation Corrections	false
	Limit Acoustic-CFL Option	Per-Model
	Secondary Gradients	On
	Convection	2nd-order
	Delta-V Dissipation	Off
Segregated Fluid Isothermal	Continuum Temperature	300.0 K
Solution Interpolation	Per Part Mapping	false
	Interpolation Method	Nearest neighbor
	Legacy Method	false
	Conservation Correction	Disable

SST (Menter) K-Omega	Curvature Correction Option	Off
	Realizability Option	Durbin Scale Limiter
	Compressibility Correction	true
	Low Re Damping Modification	false
	Convection	2nd-order
	Normal Stress Term	false
	Tke Minimum	1.0E-10
	Sdr Minimum	1.0E-10
	Secondary Gradients	On
	Kappa	0.41
	BetaStar	0.09
	Beta1	0.075
	Sigma_k1	0.85
	Sigma_w1	0.5
	Beta2	0.0828
	Sigma_k2	1.0
	Sigma_w2	0.856
	Constitutive Option	Linear
	a1	0.31
	Zeta_Star	1.5
Compressibility Parameters		
Realizability Coefficient	Realizability Coefficient	0.6
Steady	Continuum Iteration	103
Three Dimensional		
Turbulent		
Wall Distance	Wall Distance Method	Implicit Tree
Reference Values		
Minimum Allowable Wall Distance	Value	1.0E-6 m
Maximum Allowable Absolute Pressure	Value	1.0E8 Pa
Minimum Allowable Temperature	Value	5.0 C
Maximum Allowable Temperature	Value	20.0 C
Minimum Allowable Absolute Pressure	Value	1000.0 Pa
Reference Pressure	Value	101325.0 Pa
Initial Conditions		
Pressure	Method	Constant
Constant	Value	0.0 Pa
Turbulence Intensity	Method	Constant
Constant	Value	0.1
Turbulence Specification	Method	Intensity + Viscosity Ratio
Turbulent Velocity Scale	Method	Constant
Constant	Value	6.0 m/s
Turbulent Viscosity Ratio	Method	Constant
Constant	Value	10.0
Velocity	Method	Constant
	Coordinate System	Laboratory
Constant	Value	[1.0, 0.0, 0.0] m/s
Regions	Part Selection Priority	[Region]
	Regions	1

Region	Index	0
	Mesh Continuum	Parts Meshes
	Physics Continuum	Physics 1
	Parts	[Subtract]
	Type	Fluid Region
	Allow Per-Part Values	false
	Tags	[]
Boundaries	Part Surface Selection Priority	[Region: Subtract.SOLID1.Hull, Region: Subtract.SOLID1.Fin, Region: Subtract.SOLID1.Yaw, Region: Subtract.SOLID1.Fin 2, Region: Subtract.Body 2.Top&Bottom, Region: Subtract.Body 2.Outlet, Region: Subtract.Body 2.Inlet, Region: Subtract.Body 2.Sides]
	Boundaries	8
Subtract.Body 2.Inlet	Index	8
	Interfaces	
	Part Surfaces	[Subtract.Body 2.Inlet]
	Type	Velocity Inlet
	Allow Per-Surface Values	false
	Tags	[]
Physics Conditions		
Flow Direction Specification	Method	Boundary-Normal
Reference Frame Specification	Option	Lab Frame
Turbulence Specification	Method	Intensity + Viscosity Ratio
Velocity Specification	Method	Magnitude + Direction
Physics Values		
Turbulence Intensity	Method	Constant
Constant	Value	1.0
Turbulent Viscosity Ratio	Method	Constant
Constant	Value	10.0
Velocity Magnitude	Method	Constant
Constant	Value	3.0 m/s
Subtract.Body 2.Outlet	Index	7
	Interfaces	
	Part Surfaces	[Subtract.Body 2.Outlet]
	Type	Pressure Outlet
	Allow Per-Surface Values	false
	Tags	[]
Physics Conditions		
Backflow Specification	Direction	Boundary-Normal
	Pressure	Environmental
	Scalars	Specified
Pressure Outlet Option	Option	None
Reference Frame Specification	Option	Lab Frame
Turbulence Specification	Method	Intensity + Viscosity Ratio

Physics Values		
Pressure	Method	Constant
Constant	Value	0.0 Pa
Turbulence Intensity	Method	Constant
Constant	Value	0.01
Turbulent Viscosity Ratio	Method	Constant
Constant	Value	10.0
Subtract.Body 2.Sides	Index	9
	Interfaces	
	Part Surfaces	[Subtract.Body 2.Sides]
	Type	Symmetry Plane
	Allow Per-Surface Values	false
	Tags	[]
Subtract.Body 2.Top&Bottom	Index	6
	Interfaces	
	Part Surfaces	[Subtract.Body 2.Top&Bottom]
	Type	Symmetry Plane
	Allow Per-Surface Values	false
	Tags	[]
Subtract.SOLID1.Fin	Index	3
	Interfaces	
	Part Surfaces	[Subtract.SOLID1.Fin]
	Type	Wall
	Allow Per-Surface Values	false
	Tags	[]
Physics Conditions		
Reference Frame Specification	Option	Region Reference Frame
Shear Stress Specification	Method	No-Slip
Tangential Velocity Specification	Method	Fixed
Wall Surface Specification	Method	Smooth
Physics Values		
Blended Wall Function	E	9.0
	Kappa	0.42
Subtract.SOLID1.Fin 2	Index	5
	Interfaces	
	Part Surfaces	[Subtract.SOLID1.Fin 2]
	Type	Wall
	Allow Per-Surface Values	false
	Tags	[]
Physics Conditions		
Reference Frame Specification	Option	Region Reference Frame
Shear Stress Specification	Method	No-Slip
Tangential Velocity Specification	Method	Fixed
Wall Surface Specification	Method	Smooth
Physics Values		
Blended	E	9.0

Wall Function		
	Kappa	0.42
Subtract.SOLID1.Hull	Index	2
	Interfaces	
	Part Surfaces	[Subtract.SOLID1.Hull]
	Type	Wall
	Allow Per-Surface Values	false
	Tags	[]
Physics Conditions		
Reference Frame Specification	Option	Region Reference Frame
Shear Stress Specification	Method	No-Slip
Tangential Velocity Specification	Method	Fixed
Wall Surface Specification	Method	Smooth
Physics Values		
Blended Wall Function	E	9.0
	Kappa	0.42
Subtract.SOLID1.Yaw	Index	4
	Interfaces	
	Part Surfaces	[Subtract.SOLID1.Yaw]
	Type	Wall
	Allow Per-Surface Values	false
	Tags	[]
Physics Conditions		
Reference Frame Specification	Option	Region Reference Frame
Shear Stress Specification	Method	No-Slip
Tangential Velocity Specification	Method	Fixed
Wall Surface Specification	Method	Smooth
Physics Values		
Blended Wall Function	E	9.0
	Kappa	0.42
Feature Curves	Part Curve Selection Priority	[Region:Feature Curve]
	Feature Curves	1
Feature Curve	Part Curves	[Subtract.SOLID1.Default, Subtract.Body 2.Default, Subtract.Intersection]
	Tags	[]
Physics Conditions		
Initial Condition Option	Option	Use Continuum Values
Mass Source Option	Mass Source Option	false
Momentum Source Option	Momentum Source Option	None
Turbulence Source Option	Turbulence Source Option	None
Physics Values		
Axis	Coordinate System	Laboratory
	Origin	[0.0, 0.0, 0.0] m

	Direction	[0.0, 0.0, 1.0]
Motion Specification	Motion	Stationary
	Reference Frame	Lab Reference Frame
Representations		
Latest Surface/Volume	Representation	Volume Mesh
	Tags	[]
Geometry	Tags	[]
Automated Mesh.Remesh	Tags	[]
Latest Surface	Tags	[]
Volume Mesh	Cells	364132
	Interior Faces	1083256
	Vertices	422952
	Tags	[]
Finite Volume Regions		
Region	Cells	364132
	Interior Faces	1083256
	Vertices	422952
	Edges	0
Finite Volume Boundaries		
Subtract.Body 2.Inlet	Faces	364
Subtract.Body 2.Outlet	Faces	364
Subtract.Body 2.Sides	Faces	28598
Subtract.Body 2.Top&Bottom	Faces	2240
Subtract.SOLID1.Fin	Faces	3010
Subtract.SOLID1.Fin 2	Faces	4696
Subtract.SOLID1.Hull	Faces	25342
Subtract.SOLID1.Yaw	Faces	5275
Cell Sets		
Contacts		
Parts		
Body 2	Metadata	{}
	Index	2
	Region	[]
	Contacts	[]
	Descriptions	[Root]
	Face Count	12
	Tags	[]
Surfaces		
Inlet	Index	4
	Metadata	{}
	Boundary	[]
	Tags	[]
Outlet	Index	3
	Metadata	{}
	Boundary	[]

		Tags	[]
	Sides	Index	5
		Metadata	{}
		Boundary	[]
		Tags	[]
	Top&Bottom	Index	2
		Metadata	{}
		Boundary	[]
		Tags	[]
	Curves		
	Default	Index	2
		Feature Curve	[]
		Tags	[]
Cylinder 2		Metadata	{}
		Index	6
		Region	[]
		Contacts	[]
		Descriptions	[Root]
		Face Count	356
		Coordinate System	Laboratory
		Start Coordinate	[4.0, 0.0, 0.3] m,m,m
		End Coordinate	[4.0, 0.0, 0.05] m,m,m
		Radius	0.04 m
		Tags	[]
	Surfaces		
	Cylinder Surface	Index	42
		Metadata	{}
		Boundary	[]
		Tags	[]
	Curves		
	Cylinder Curve	Index	10
		Feature Curve	[]
		Tags	[]
Cylinder 3		Metadata	{}
		Index	7
		Region	[]
		Contacts	[]
		Descriptions	[Root]
		Face Count	356
		Coordinate System	Laboratory
		Start Coordinate	[4.0, 0.0, -0.3] m,m,m
		End Coordinate	[4.0, 0.0, 0.0] m,m,m
		Radius	0.04 m
		Tags	[]
	Surfaces		
	Cylinder Surface	Index	43
		Metadata	{}
		Boundary	[]
		Tags	[]
	Curves		
	Cylinder Curve	Index	11
		Feature Curve	[]
		Tags	[]
Cylinder 4		Metadata	{}
		Index	8
		Region	[]
		Contacts	[]
		Descriptions	[Root]
		Face Count	180

		Coordinate System	Laboratory
		Start Coordinate	[4.04, 0.25, 0.0] m,m,m
		End Coordinate	[3.8, 0.25, 0.0] m,m,m
		Radius	0.03 m
		Tags	[]
	Surfaces		
	Cylinder Surface	Index	44
		Metadata	{}
		Boundary	[]
		Tags	[]
	Curves		
	Cylinder Curve	Index	12
		Feature Curve	[]
		Tags	[]
fin		Metadata	{}
		Index	5
		Region	[]
		Contacts	[]
		Descriptions	[Root]
		Face Count	356
		Coordinate System	Laboratory
		Start Coordinate	[1.3, 0.0, 0.5] m,m,m
		End Coordinate	[1.3, 0.0, 0.2] m,m,m
		Radius	0.05 m
		Tags	[]
	Surfaces		
	Cylinder Surface	Index	41
		Metadata	{}
		Boundary	[]
		Tags	[]
	Curves		
	Cylinder Curve	Index	9
		Feature Curve	[]
		Tags	[]
SOLID1		Metadata	{}
		Index	1
		Region	[]
		Contacts	[]
		Descriptions	[Root]
		Face Count	77476
		Tags	[]
	Surfaces		
	Fin	Index	6
		Metadata	{}
		Boundary	[]
		Tags	[]
	Fin 2	Index	31
		Metadata	{}
		Boundary	[]
		Tags	[]
	Hull	Index	1
		Metadata	{}
		Boundary	[]
		Tags	[]
	Yaw	Index	14
		Metadata	{}
		Boundary	[]
		Tags	[]
	Curves		

Default	Index	1
	Feature Curve	[]
	Tags	[]
Subtract	Metadata	{}
	Index	4
	Region	[Region]
	Contacts	[]
	Descriptions	[Root, Automated Mesh.Remesh]
	Face Count	3292
	Operation	Subtract
	Tags	[]
3D-CAD Models		
3D-CAD Model 1	Part Update Method	UPDATE_GEOMETRY
	Tags	[]
Body Groups		
Body 2	Name	Body 2
	Color	java.awt.Color[r=128,g=128,b=128]
	Opacity	1.0
	Tags	[]
SOLID1	Name	SOLID1
	Color	java.awt.Color[r=128,g=128,b=128]
	Opacity	1.0
	Tags	[]
Features		
XY	Error Message	
	Origin	[0.0, 0.0, 0.0] m
	X-Axis	[1.0, 0.0, 0.0]
	Y-Axis	[0.0, 1.0, 0.0]
	Tags	[]
YZ	Error Message	
	Origin	[0.0, 0.0, 0.0] m
	X-Axis	[0.0, 1.0, 0.0]
	Y-Axis	[0.0, 0.0, 1.0]
	Tags	[]
ZX	Error Message	
	Origin	[0.0, 0.0, 0.0] m
	X-Axis	[0.0, 0.0, 1.0]
	Y-Axis	[1.0, 0.0, 0.0]
	Tags	[]
Global Origin	Error Message	
	Position	[0.0, 0.0, 0.0]
	Tags	[]
Lab Coordinate System	Error Message	
	Origin	[0.0, 0.0, 0.0]
	X-axis Direction	[1.0, 0.0, 0.0]
	Y-axis Direction	[0.0, 1.0, 0.0]
	Tags	[]
ImportCad 1	Error Message	
	Exchange Settings	{STEP=0, NX=0, CATIAV5=0, SE=0, JT=0}
	File Name	Z:\VIVO\DarpaSuboffBaseline_igs.prt
	Tags	[]
Sketch 1	Error Message	
	Tags	[]
Extrude 1	Error Message	
	Sketch	Sketch 1
	Method	Blind
	Direction Type	Normal
	Extrusion Options	OneWay
	Distance	5.0 m

	Asym. Distance	0.1 m
	Draft	None
	Draft Angle	10.0 deg
	Offset Distance	0.1 m
	Body Type	Solid
	Body Interaction	None
	Bodies To Interact	All
	Tags	[]
Design Filters		
Design Parameters		
Operations		
Subtract	Link Output Part Name	false
	Input Parts	[SOLID1, Body 2]
	Target Part	[Body 2]
	Perform CAD Boolean	true
	Output Parts	[Subtract]
	Tags	[]
Tessellation Options	Tessellation Options	Medium
Tolerance Type	Tolerance Type	Precise
	Tolerance	1.0E-5 m
Automated Mesh	Per-Part Meshing	false
	Mesher Execution Mode	Serial
	Input Parts	[Subtract]
	Perform Local Surface Meshing	false
	Preserve Surface Perimeters	None
	Verbose Output	false
	Tags	[]
Meshers		
Surface Remesher	Meshing Method	Enhanced Quality Triangle
	Perform Curvature Refinement	true
	Perform Proximity Refinement	true
	Perform Compatibility Refinement	false
	Create Aligned Meshes	true
	Minimum Face Quality	0.01
Trimmed Cell Mesher	Core Cell Type	Hexahedra
	Coordinate System	Laboratory
	Perform Mesh Alignment	false
	Size Dependent Growth Rates	All cells
	Run Post Mesh Optimizer	true
Prism Layer Mesher	Stretching Function	Geometric Progression
	Distribution Mode	Stretch Factor
	Boundary March Angle	25.0
	Gap Fill Percentage	25.0
	Minimum Thickness Percentage	10.0
	Layer Reduction Percentage	50.0
	Concave Angle Limit	0.0
	Convex Angle Limit	360.0
	Near Core Layer Aspect Ratio	0.5
Default Controls		
Base Size	Base Size	0.1 m
	Base Size	0.1 m
CAD Projection	Project to CAD	true
Target Surface Size	Size Type	Relative to base
	Percentage of Base	100.0
	Absolute Size	0.1 m
Minimum Surface Size	Size Type	Relative to base

Curvature	Surface	Percentage of Base	25.0
		Absolute Size	0.025 m
		Enable Curvature Deviation Distance	false
		# Pts/circle	76.0
		Max # Pts/circle	200.0
		Curvature Deviation Distance	0.01 m
	Surface Proximity	Search Floor	0.0 m
		# Points in gap	2.0
		Enable Search Ceiling	false
		Search Ceiling	1.0E10 m
	Surface Growth Rate	Surface Growth Rate	SLOW
		User Specified Value	1.3
	Number of Prism Layers	Number of Prism Layers	6
	Prism Layer Stretching	Prism Layer Stretching	1.0
	Prism Layer Total Thickness	Size Type	Relative to base
		Percentage of Base	25.0
		Absolute Size	0.025 m
	Maximum Core/Prism Transition Ratio	Limit Cell Size by Prism Layer Thickness	false
	Volume Growth Rate	Default Growth Rate	Medium
		Surface Growth Rate	Disable
	Maximum Cell Size	Size Type	Relative to base
		Percentage of Base	10000.0
		Absolute Size	10.0 m
	Post Mesh Optimization	Optimize Boundary Vertices	false
		Optimize Cell Topology	false
Custom Controls			
fin		Enable Control	true
		Controls Display Mode	All
		Part Surfaces	[Subtract.SOLID1.Fin]
		Apply Only to Contacting Area	false
		Tags	[]
Controls			
Target Surface Size		Target Surface Size	Custom
Minimum Surface Size		Minimum Surface Size	Custom
Surface Curvature		Curvature	Parent
Surface Proximity		Proximity	Parent
Edge Proximity		Proximity	Parent
Surface Growth Rate		Surface Growth Rate	Parent
Surface Remeshing		Surface Remeshing	Parent
Meshing Method		Meshing Method	Parent
Prism Layers		Prism Layers	Parent
Trimmer Surface Growth Rate		Trimmer Surface Growth Rate	Parent
Remesher Wake Refinement		Specify wake refinement options	false

	Trimmer Wake Refinement	Specify Trimmer Wake Refinement Options	false
	Values		
	Target Surface Size	Size Type	Relative to base
		Percentage of Base	15.0
		Absolute Size	0.015 m
	Minimum Surface Size	Size Type	Relative to base
		Percentage of Base	8.0
		Absolute Size	0.008 m
fin 2		Enable Control	true
		Controls Display Mode	All
		Parts	[fin]
		Tags	[]
	Controls		
	Surface Remesher	Customize Size	true
	Prism Layer Mesher	Customize Number of Layers	false
		Customize Total Thickness	true
		Customize Stretching	true
	Trimmer	Customize Isotropic Size	false
		Customize Anisotropic Size	false
	Values		
	Custom Size	Size Type	Relative to base
		Percentage of Base	3.0
		Absolute Size	0.0030000000000000005 m
	Custom Prism Values		
	Prism Layer Stretching	Prism Layer Stretching	1.0
	Prism Layer Total Thickness	Size Type	Relative to base
		Percentage of Base	25.0
		Absolute Size	0.025 m
hull		Enable Control	true
		Controls Display Mode	All
		Part Surfaces	[Subtract.SOLID1.Hull]
		Apply Only to Contacting Area	false
		Tags	[]
	Controls		
	Target Surface Size	Target Surface Size	Custom
	Minimum Surface Size	Minimum Surface Size	Custom
	Surface Curvature	Curvature	Parent
	Surface Proximity	Proximity	Parent
	Edge Proximity	Proximity	Parent
	Surface Growth Rate	Surface Growth Rate	Parent
	Surface Remeshing	Surface Remeshing	Parent
	Meshing Method	Meshing Method	Parent
	Prism Layers	Prism Layers	Parent
	Trimmer	Trimmer Surface Growth Rate	Parent

Surface Growth Rate			
	Remesher	Specify wake refinement options	false
Wake Refinement			
	Trimmer	Specify Trimmer Wake Refinement Options	false
Wake Refinement			
	Values		
	Target	Size Type	Relative to base
Surface Size			
		Percentage of Base	15.0
		Absolute Size	0.015 m
	Minimum	Size Type	Relative to base
Surface Size			
		Percentage of Base	8.0
		Absolute Size	0.008 m
pitch		Enable Control	true
		Controls Display Mode	All
		Part Surfaces	[Subtract.SOLID1.Fin 2]
		Apply Only to Contacting Area	false
		Tags	[]
	Controls		
	Target	Target Surface Size	Custom
Surface Size			
	Minimum	Minimum Surface Size	Custom
Surface Size			
	Surface	Curvature	Parent
Curvature			
	Surface	Proximity	Parent
Proximity			
	Edge	Proximity	Parent
Proximity			
	Surface	Surface Growth Rate	Parent
Growth Rate			
	Surface	Surface Remeshing	Parent
Remeshing			
	Meshing	Meshing Method	Parent
Method			
	Prism	Prism Layers	Parent
Layers			
	Trimmer	Trimmer Surface Growth Rate	Parent
Surface Growth Rate			
	Remesher	Specify wake refinement options	false
Wake Refinement			
	Trimmer	Specify Trimmer Wake Refinement Options	false
Wake Refinement			
	Values		
	Target	Size Type	Relative to base
Surface Size			
		Percentage of Base	6.0
		Absolute Size	0.006000000000000001 m
	Minimum	Size Type	Relative to base
Surface Size			
		Percentage of Base	5.0
		Absolute Size	0.005 m
Surface Control		Enable Control	true
		Controls Display Mode	All
		Part Surfaces	[Subtract.Body 2.Inlet, Subtract.Body 2.Outlet, Subtract.Body 2.Sides, Subtract.Body 2.Top&Bottom]
		Apply Only to Contacting Area	false
		Tags	[]
	Controls		

Surface Size	Target	Target Surface Size	Custom
Surface Size	Minimum	Minimum Surface Size	Custom
Curvature	Surface	Curvature	Parent
Proximity	Surface	Proximity	Parent
Proximity	Edge	Proximity	Parent
Growth Rate	Surface	Surface Growth Rate	Parent
Remeshing	Surface	Surface Remeshing	Parent
Method	Meshing	Meshing Method	Parent
Layers	Prism	Prism Layers	Parent
Surface Growth Rate	Trimmer	Trimmer Surface Growth Rate	Parent
Wake Refinement	Remesher	Specify wake refinement options	false
Wake Refinement	Trimmer	Specify Trimmer Wake Refinement Options	false
	Values		
Surface Size	Target	Size Type	Absolute
		Percentage of Base	500.0
		Absolute Size	0.5 m
Surface Size	Minimum	Size Type	Absolute
		Percentage of Base	20.0
		Absolute Size	0.02 m
yaw		Enable Control	true
		Controls Display Mode	All
		Part Surfaces	[Subtract.SOLID1.Yaw]
		Apply Only to Contacting Area	false
		Tags	[]
	Controls		
Surface Size	Target	Target Surface Size	Custom
Surface Size	Minimum	Minimum Surface Size	Custom
Curvature	Surface	Curvature	Parent
Proximity	Surface	Proximity	Parent
Proximity	Edge	Proximity	Parent
Growth Rate	Surface	Surface Growth Rate	Parent
Remeshing	Surface	Surface Remeshing	Parent
Method	Meshing	Meshing Method	Parent
Layers	Prism	Prism Layers	Parent
Surface Growth Rate	Trimmer	Trimmer Surface Growth Rate	Parent
Wake Refinement	Remesher	Specify wake refinement options	false
Wake Refinement	Trimmer	Specify Trimmer Wake Refinement Options	false
	Values		

Target Surface Size	Size Type	Relative to base
	Percentage of Base	6.0
	Absolute Size	0.006000000000000001 m
Minimum Surface Size	Size Type	Relative to base
	Percentage of Base	5.0
	Absolute Size	0.005 m
yaw&pitch	Enable Control	true
	Controls Display Mode	All
	Parts	[Cylinder 2, Cylinder 3, Cylinder 4]
	Tags	[]
Controls		
Surface Remesher	Customize Size	true
Prism Layer Mesher	Customize Number of Layers	false
	Customize Total Thickness	true
	Customize Stretching	true
Trimmer	Customize Isotropic Size	false
	Customize Anisotropic Size	false
Values		
Custom Size	Size Type	Relative to base
	Percentage of Base	3.0
	Absolute Size	0.0030000000000000005 m
Custom Prism Values		
Prism Layer Stretching	Prism Layer Stretching	1.0
Prism Layer Total Thickness	Size Type	Relative to base
	Percentage of Base	25.0
	Absolute Size	0.025 m
Descriptions	Number of Children	4
Root	Described Parts	[SOLID1, Body 2, Subtract, fin, Cylinder 2, Cylinder 3, Cylinder 4]
Automated Mesh.Remesh	Described Parts	[Subtract]
	Faces	111292
	Vertices	55648
Latest Surface	Described Parts	[SOLID1, Body 2, Subtract, fin, Cylinder 2, Cylinder 3, Cylinder 4]
	Faces	190028
	Vertices	95028
	Preview Mesh Operation Parts	false
Latest Surface/Volume	Described Parts	[SOLID1, Body 2, Subtract, fin, Cylinder 2, Cylinder 3, Cylinder 4]
Coordinate Systems		
Laboratory	Tags	[]
Local Coordinate Systems		
Tables	Tables	0
Tags		
Units	Preferred System	Systeme Internationale
Custom Trees		
Filters		
Field Functions		
drag	Function Name	drag force
	Inverse Distance Weight	false
	Value Type	Scalar
	Assembly Code	${Force1Report}

	Definition	${Force1Report}
	Ignore Boundary Values	false
	Tags	[]
Simulation Operations	Selected	[]
Parameters		
Volume Shapes		
Data Set Functions	Data Directory	function_data
Update Events	Event Count	0
	Event Names	
Time Scales		
User Code		
Data Focus		
Layouts		
default		
Data Mappers	Use Legacy Volume Mapper	false
Motions		
Stationary	Tags	[]
Reference Frames		
Lab Reference Frame	Tags	[]
Screenplays		
Derived Parts	Derived Parts	1
Plane Section	Origin	[4.198330850483278, 2.4533522797891845, -5.245747083234796E-4] m,m,m
	Coordinate System	Laboratory
	Normal	[0.0, 0.0, 1.0] m,m,m
	Parts	[Region]
	Section Mode	SINGLE
	Displayed Index	-1
	Tags	[]
Single section	Offset	0.0 m
Summaries		
Monitors	Monitors	11
	Monitors To Print	[Continuity, X-momentum, Y-momentum, Z-momentum, Tke, Sdr, Energy, Force 1 Monitor, Force 1 Monitor 2]
	Output Direction	Horizontal
	Heading Print Frequency	10
Force 1 Monitor	Report	[Drag]
	Enabled	true
	Value Type	Total Value
	Trigger	Iteration
	Normalization Option	Off
	Maximum Plot Samples	5000
	Tags	[]
Iteration Frequency	Iteration Frequency	1
	Start Iteration	0
	Enable Stop	false
	Stop Iteration	0
Force 1 Monitor 2	Report	[Force 1]
	Enabled	true
	Value Type	Total Value
	Trigger	Iteration
	Normalization Option	Off
	Maximum Plot Samples	5000
	Tags	[]
Iteration Frequency	Iteration Frequency	1
	Start Iteration	0
	Enable Stop	false

	Stop Iteration	0
Iteration	Maximum Plot Samples	5000
	Tags	[]
Physical Time	Maximum Plot Samples	5000
	Tags	[]
Reports	Reports	2
Drag	Units	N
	Coordinate System	Laboratory
	Direction	[1.0, 0.0, 0.0]
	Force Option	Pressure + Shear
	Reference Pressure	0.0 Pa
	Number of Bands	0
	Parts	[Region: Subtract.SOLID1.Fin, Region: Subtract.SOLID1.Fin 2, Region: Subtract.SOLID1.Hull, Region: Subtract.SOLID1.Yaw]
	Representation	Volume Mesh
	Smooth Values	false
	Tags	[]
Force 1	Units	N
	Coordinate System	Laboratory
	Direction	[1.0, 0.0, 0.0]
	Force Option	Pressure + Shear
	Reference Pressure	101325.0 Pa
	Number of Bands	0
	Parts	[Region: Subtract.SOLID1.Fin, Region: Subtract.SOLID1.Fin 2, Region: Subtract.SOLID1.Hull, Region: Subtract.SOLID1.Yaw]
	Representation	Volume Mesh
	Smooth Values	false
	Tags	[]
Solvers		
Steady		
Stopping Criteria	Verbose	false
Partitioning	Solver Frozen	false
	Partitioning Method	Per-Continuum
Wall Distance	Solver Frozen	false
	Verbosity	0
	Minimum Tree Size Threshold	500000
Segregated Flow	Pressure Gradient Inverse Distance Weighting	true
	Freeze Flow	false
	Reconstruction Frozen	false
	Reconstruction Zeroed	false
	Enable Enhanced Stability Treatment	false
	Temporary Storage Retained	false
	Pressure Corrections: Bad Cell Minimum Scaling	0.8
	Pressure Corrections: Acceptable Cell Volume Change	0.001
	Velocity Corrections: Maximum Unlimited Velocity	20.0 m/s
	Velocity Corrections: Acceptable Velocity Increase Rate [<1]	0.15
	Continuity Initialization	false
Velocity	Under-Relaxation Factor	0.7
	Dynamic Local Under-Relaxation	false
Under-Relaxation Factor Ramp	Ramp Method	No Ramp
AMG Linear	Max Cycles	30

Solver			
		Verbosity	NONE
		Enable Direct Solver	false
		Maximum Direct Solver Equations	32
		Convergence Tolerance	0.1
		Epsilon	0.0
		Cycle Type	Flex Cycle
		Group Size Control	Auto
		Group Size	4
		Relaxation Scheme	Gauss-Seidel
		Acceleration Method	None
		Scaling	Disabled
	Flex Cycle	Restriction Tolerance	0.9
		Prolongation Tolerance	0.5
		Sweeps	1
	Pressure	Number of Non-Orthogonality Correctors	0
		Under-Relaxation Factor	0.3
		Pressure Reference Location	Automatic Selection
	Under-Relaxation Factor Ramp	Ramp Method	No Ramp
	AMG Linear Solver	Max Cycles	30
		Verbosity	NONE
		Enable Direct Solver	false
		Maximum Direct Solver Equations	32
		Convergence Tolerance	0.1
		Epsilon	0.0
		Cycle Type	V Cycle
		Group Size Control	Auto
		Group Size	4
		Relaxation Scheme	Gauss-Seidel
		Acceleration Method	Bi Conjugate Gradient Stabilized
		Scaling	Auto
	V Cycle	Pre-Sweeps	1
		Post-Sweeps	1
		Max Levels	50
	Segregated Energy	Solver Frozen	false
		Reconstruction Frozen	false
		Reconstruction Zeroed	false
		Temporary Storage Retained	false
		Fluid Under-Relaxation Factor	0.9
		Solid Under-Relaxation Factor	0.99
		Enable High-Accuracy Temporal Discretization	false
	Fluid Under-Relaxation Factor Ramp	Ramp Method	No Ramp
	Solid Under-Relaxation Factor Ramp	Ramp Method	No Ramp
	AMG Linear Solver	Max Cycles	30
		Verbosity	NONE
		Enable Direct Solver	false
		Maximum Direct Solver Equations	32
		Convergence Tolerance	0.1
		Epsilon	0.0
		Cycle Type	V Cycle
		Group Size Control	Auto
		Group Size	4
		Relaxation Scheme	Gauss-Seidel
		Acceleration Method	None
		Scaling	Disabled

V Cycle	Pre-Sweeps	1
	Post-Sweeps	1
	Max Levels	50
K-Omega Turbulence	Solver Frozen	false
	Reconstruction Frozen	false
	Reconstruction Zeroed	false
	Temporary Storage Retained	false
	Under-Relaxation Factor	0.8
	Boundary Layer Initialization	false
Under-Relaxation Factor Ramp	Ramp Method	No Ramp
AMG Linear Solver	Max Cycles	30
	Verbosity	NONE
	Enable Direct Solver	false
	Maximum Direct Solver Equations	32
	Convergence Tolerance	0.1
	Epsilon	0.0
	Cycle Type	Flex Cycle
	Group Size Control	Auto
	Group Size	4
	Relaxation Scheme	Gauss-Seidel
	Acceleration Method	None
	Scaling	Disabled
Flex Cycle	Restriction Tolerance	0.9
	Prolongation Tolerance	0.5
	Sweeps	1
K-Omega Turbulent Viscosity	Solver Frozen	false
	Under-Relaxation Factor	1.0
	Maximum Ratio	100000.0
Stopping Criteria	Verbose	false
Maximum Steps	Enabled	true
	Maximum Steps	200
	Logical Rule	Or
	Criterion Satisfied	false
	Tags	[]
Stop File	Enabled	true
	Stop Inner Iterations	true
	Path	ABORT
	Logical Rule	Or
	Criterion Satisfied	false
	Tags	[]
Solution Histories		
Solution Views		
Current Solution	Iteration	103
	Time Step	0
	Solution Time	0.0
	Tags	[]
Layout Views		

Solution

Accumulated CPU Time over all processes (s)	330.4859999999999
Elapsed Time (s)	165.2312389997678
Iterations	103